

SMART INDIA HACKATHON 2025

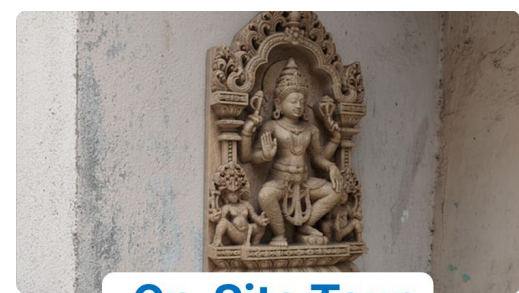


- Problem Statement ID - **SIH25052**
- Problem Statement Title - **AR-Based Cultural Heritage Preservation Platform**
- Theme - **Heritage & Culture**
- PS Category - **Software**
- Team ID - **72696**
- Team Name (Registered on portal) - **Invariants**



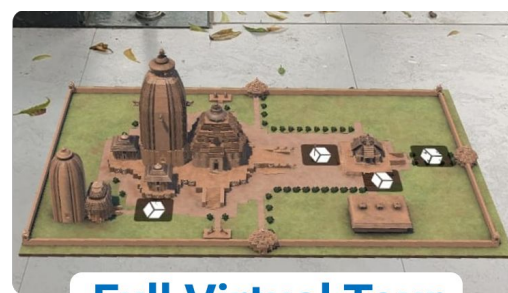
Proposed Solution

- An AR app with **3D reconstructed** and realistic compressed **GLB models** of heritage sites.
- **Dual-mode:** On-site AR overlays and immersive Virtual Tour.
- **Offline-first AR** for reliable use in rural areas, **packet split transfer** for **low-bandwidth** areas.
- **AI-driven, 20+ languages** personal guide for tailored **storytelling**.
- Local **economic boost** through **partnership**.
- **Crowd-sourced & moderated contributions** to enrich heritage records.
- Centralized **admin dashboard** for analytics, security, and content governance.



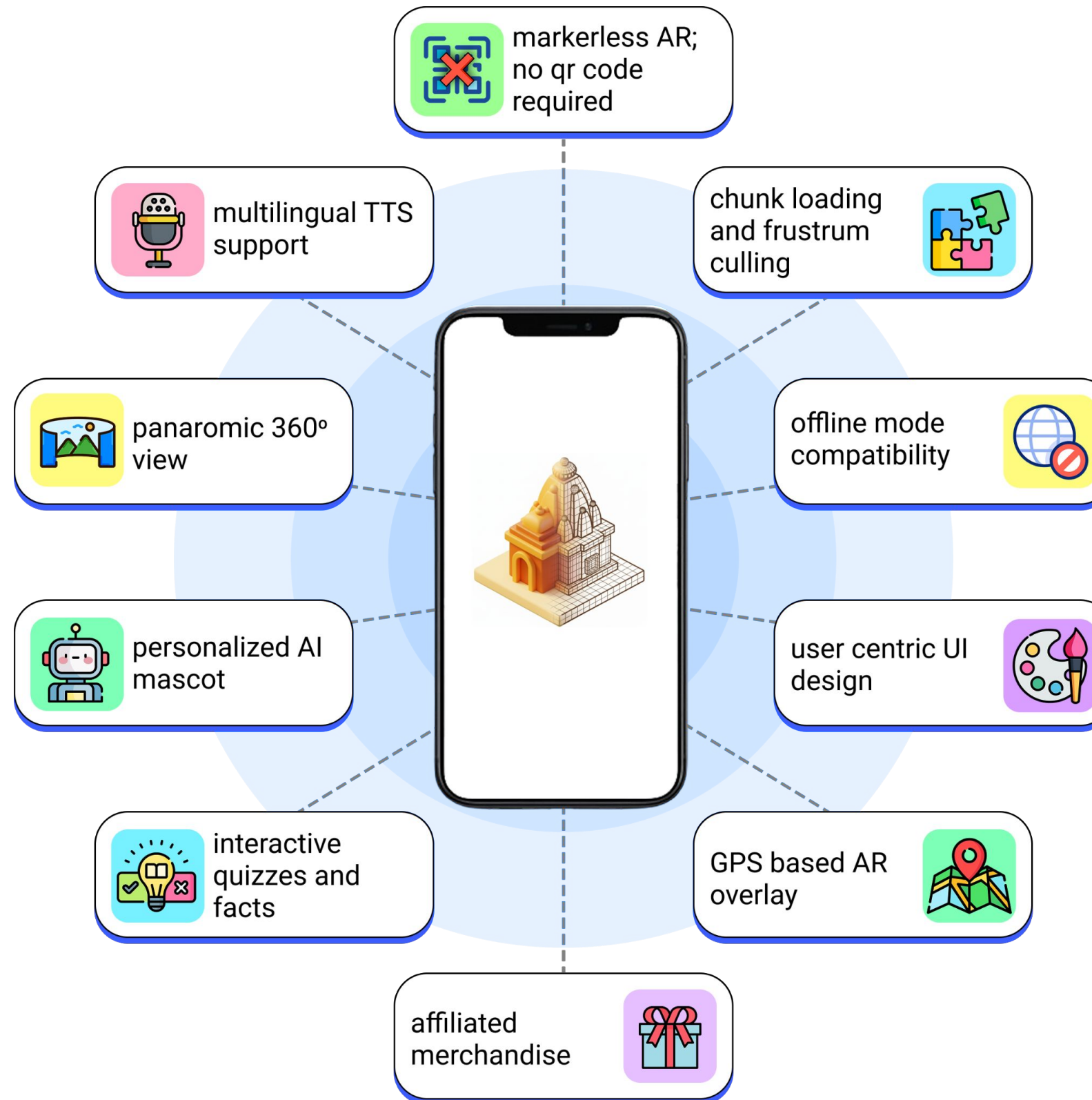
On-Site Tour

Experience history with immersive AR storytelling on location



Full Virtual Tour

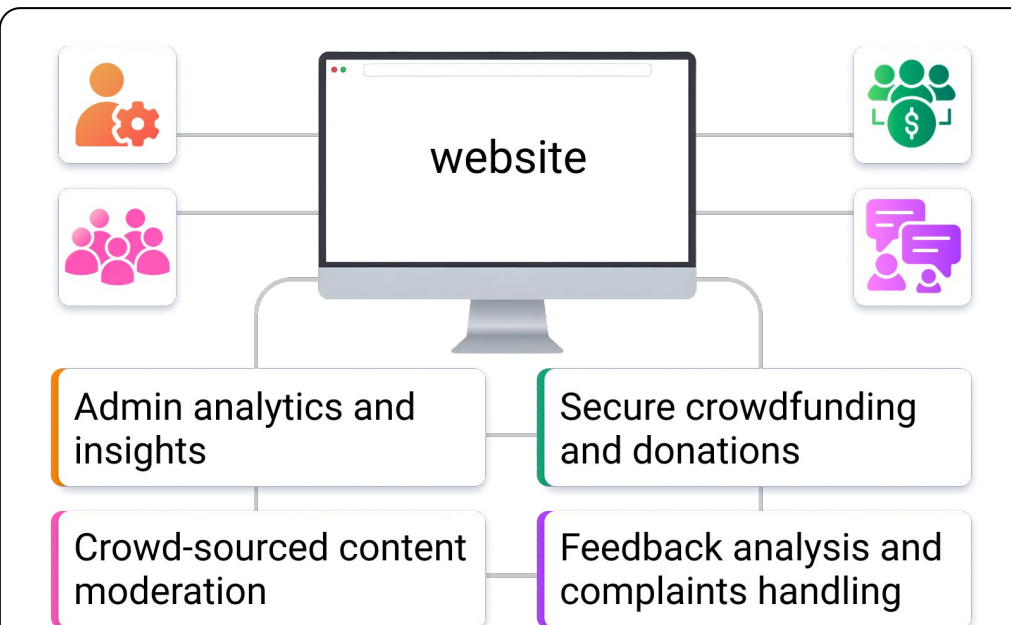
lets you explore heritage sites from home with interactive 3d model



"bringing the past back to life forever"

It Addresses the Problem by

- **Digitally preserving heritage** via accurate **3D models** resilient to **decay**.
- **Scalable & reliable** with progressive streaming & offline-first design.
- Supports **sustainable** tourism, artisans livelihood, & cultural education.
- **Builds trust** with moderated contributions & **secure** data practices.



TECHNICAL APPROACH

Tech Stack

App Frontend

Unity AR Foundation

Draco ARCore XRKit GLB Viewer

XR Interaction Kit

Website Frontend

Nextjs React

Typescript Tailwind CSS Axios

Zustand Shadcn UI OAuth Stripe

Backend (both)

Nodejs Expressjs

Javascript Python Django Prisma

APIs & AI

Google Maps Groq SDK

Gemini AI Clerk

Database / Cloud Storage

SQL

PostgreSQL Sheets Google Cloud

Deploy & Dev

Vercel Render Git

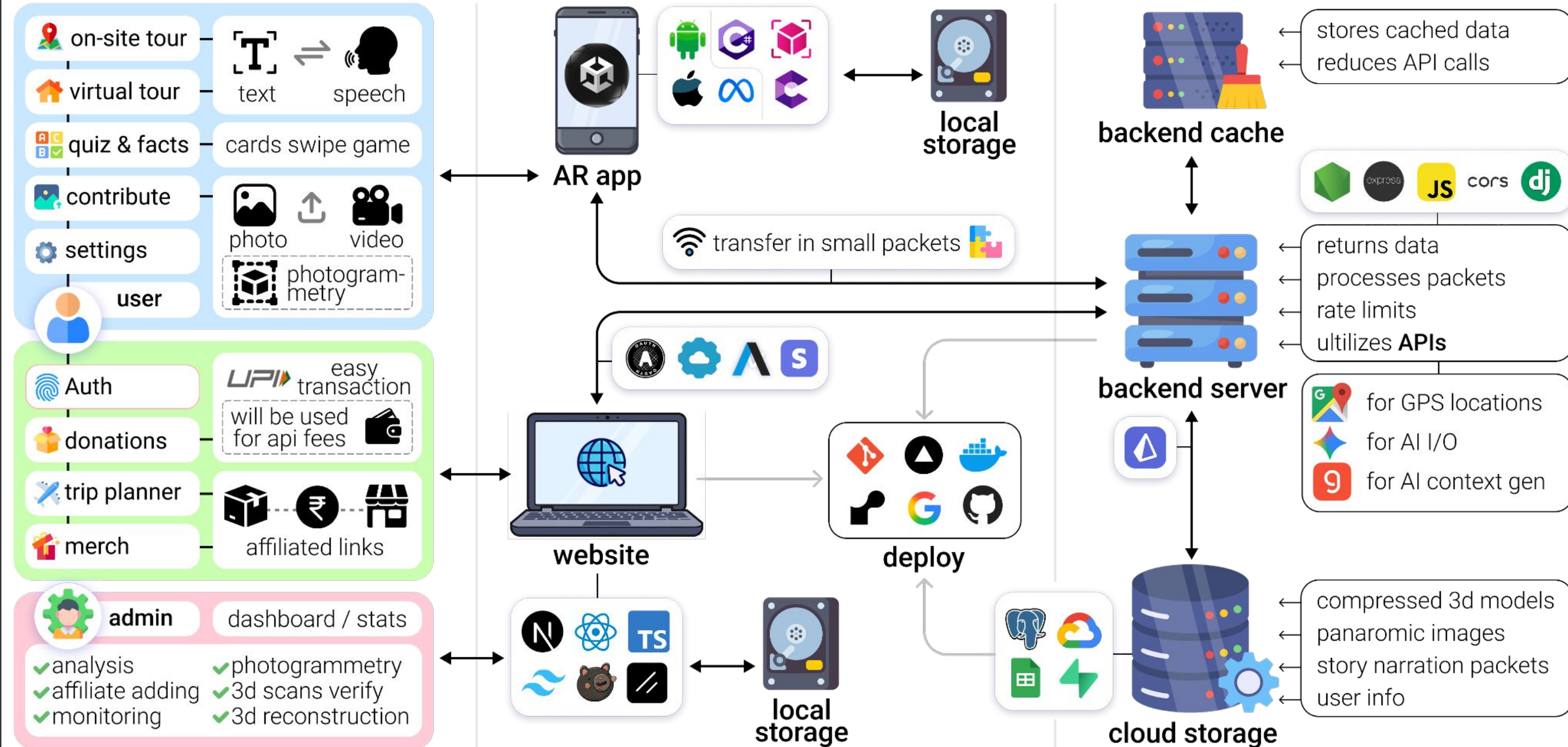
Github Docker VS Code npm

Photoshop Blender Zbrush

Modular, scalable tech stack adaptable to future frameworks

**Our Working Prototype:** [click here](#)

progress made so far: 60%



Financials



Advertisements
In-app ad revenue



Donations
Voluntary user contributions



In-app Purchases
Premium tours



Affiliated Links
Earnings from partner links



Others ₹32K - ₹38K
Data Privacy | Encryption |
Monitoring | Compliance



AR App ₹65K - ₹72K
UI/UX Design | AR SDKs |
Testing | Accessibility



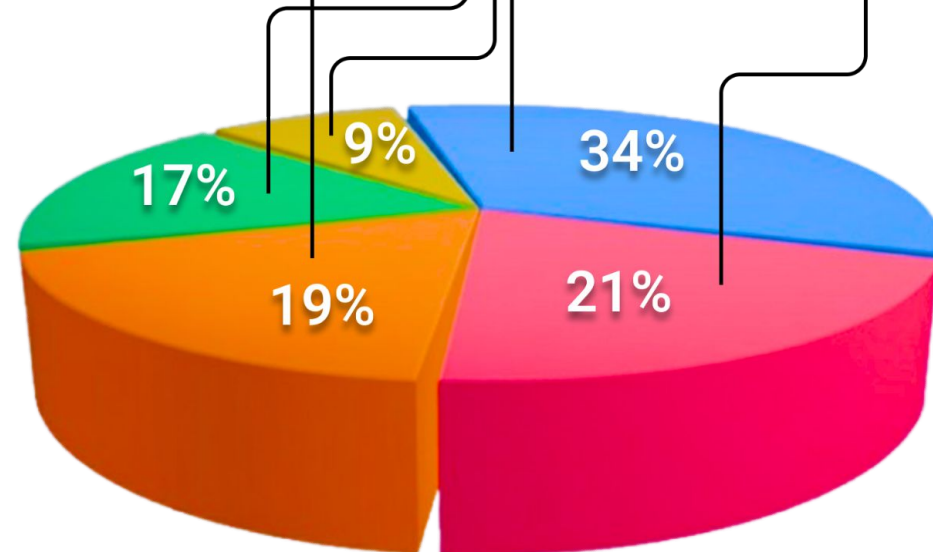
3D Models ₹1.3L - ₹1.4L
Modeling | Acquisition |
Optimization | Management



Backend ₹70K - ₹80K
APIs/DB | Authentication |
Analytics | Integration



Cloud ₹75K - ₹85K
Compute | Storage |
Bandwidth | Security



Feasibility



Proven AR tech built on **Unity, ARCore** and **ARKit** for **cross-platform** use.



Photogrammetry + AR can digitize **1,000+ monuments** with high accuracy.



Offline-first AR ensures access at **35%+ rural heritage sites** with weak connectivity.



Cloud hosting + 3D compression reduce model size by **60%**, low-bandwidth access.

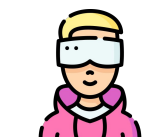
Viability



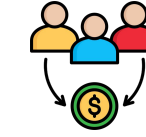
India's tourism economy to hit **\$250B by 2030**, **20% heritage-linked**.



Digital collectibles & gamified AR experiences boost tourist spend by **20%**.



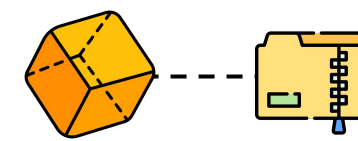
AR tools in **schools & museums** reach **40M+ students**.



Public-private partnerships with artisans enable affiliate sales, supporting local crafts.

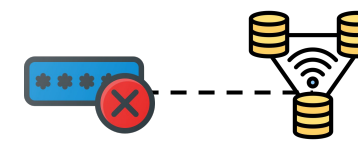
Challenges and Mitigation

Heavy 3D files slow down devices



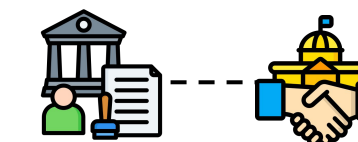
File compression using **blender + unity's glb viewer** and **draco** compression & **LOD (Level of Detail)** cuts load time by up to **60%**

Weak rural internet



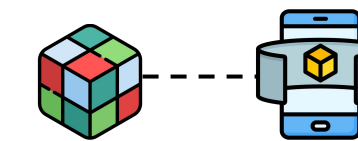
Hybrid cache/CDN may deliver content **2x faster** in low-connectivity zones

ASI/government scan approvals needed



ASI partnerships + photogrammetry accelerate **government approvals** for new AR content

Device fragmentation (Android/iOS)

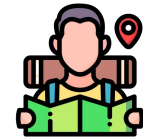


Testing aims to cover **95% of major devices**; **fallback AR** planned for **older models**

IMPACT AND BENEFITS



AR-enabled sites see a **15–20% rise in tourist engagement.**



82% of travelers prioritize cultural and heritage **AR experiences**



AR boosts knowledge **retention by 70%** vs traditional guides.



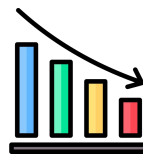
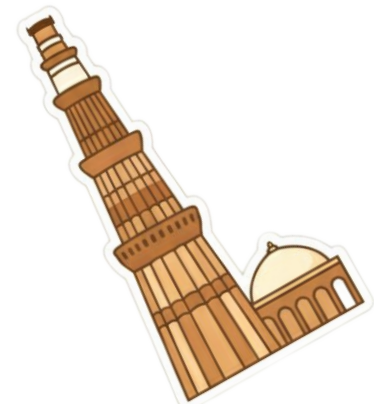
Storytelling in **20+ languages** can reach **650M+ tourists.**



Offline-first AR works in rural areas with **low connectivity**



India's AR/VR education market: **\$917M by 2025.**



Virtual tours can **cut physical strain** on sensitive sites by **15–20%.**



Heritage tourism in India projected at **\$57B by 2033.**

IMPACT



BENEFITS



Tourists: Multilingual AR tours solving info & language gaps.



Students: Interactive AR lessons boost engagement and recall by **70%.**



Local Guides: AR tools for storytelling, boosts **earnings by 30%.**



Artisans: Access a **\$2.5B+** digital marketplace, creating new income.



Government & ASI: AR preservation cuts costs **40%.**



Rural Communities: More visibility, **150K+** jobs supported.



Schools & Museums: Reach **40M+** learners with immersive edtech.



Global/NRI access, **amplifying** cultural reach.



Research Papers

AR in Cultural Heritage

- Augmented Reality in Cultural Heritage: An Overview of the Last Decade - Boboc et al., Applied Sciences, 2022 [\[Link\]](#)
- AI-driven 3D Restoration of Degraded Artefacts for Museum Digitalization - ScienceDirect, 2024 [\[Link\]](#)
- Deep Learning for Single Photo 3D Reconstruction of Cultural Heritage - ResearchGate, 2025 [\[Link\]](#)
- ItihasHub: A Digital Platform for the Preservation and Promotion of India's Cultural Heritage – Chaurasia et al., JETIR, 2025 [\[Link\]](#)

Government Applications

- National Digital Tourism Mission (NDTM) - Ministry of Tourism, Govt. of India, 2022 [\[Link\]](#)
- Preserving Indian Heritage with Digital Innovation - Kumara, Raghuvanshi & Surya, Int. Journal of Arts Architecture & Design, 2025 [\[Link\]](#)

User Demand & Community Insights

- “Best AR apps for history & museums” - r/augmentedreality, 2025 [\[Link\]](#)
- Visitors’ Acceptance of Wearable AR Technology in Museums - SAGE Journals, 2024 [\[Link\]](#)

Our Survey: [click here](#)

we conducted survey on Sept 2025
300+ respondents (Tourists,Students etc.)

Our Prototype: [click here](#)

Competitors Comparison	 SmarakaAR	 Augtraveler	 JuárezAR	 Xplore
Virtual Tour	✓	✓	✗	✓
Offline Mode	✓	✗	✓	✗
AI-Support	✓	✗	✗	✗
Multi-language	✓	✓	✗	✓
3D Reconstruction	✓	✗	✓	✓
Audio-Guide	✓	✓	✗	✓
Revenue Model	✓	✓	✗	✗

Industry Outlook

\$6.2B	AR/VR Tourism by 2030 (20% CAGR)
\$250B	India’s Tourism sector by 2030 40% heritage-linked
15–20%	Boost in engagement at AR-enabled heritage sites
70%	Higher cultural learning retention with AR
42	UNESCO Sites (India ranks 6th globally)

Primary Research(survey)

65.38%	of respondents reported a lack of information or guide support during heritage site visits.
50.00%	mentioned that high entry costs were a significant concern.
34.62%	pointed out language barriers in understanding the historical context.
30.77%	expressed that poor accessibility /connectivity made it difficult to reach or explore sites.

